# ABB EngineeredX 2.0 Industrial AutoML Report

Job ID: smoketest
Instruction: Predict motor failure and explain vibration risk
Task Type: classification
Industrial Context: predictive maintenance
Best Model: Gradient Boosting Classifier
Metrics: {'accuracy': 1.0, 'f1': 1.0, 'roc_auc': 1.0}

Leaderboard:
- Gradient Boosting Classifier: {'accuracy': 1.0, 'f1': 1.0, 'roc_auc': 1.0}
- Logistic Regression: {'accuracy': 0.9565, 'f1': 0.9353, 'roc_auc': 1.0}
- SVM: {'accuracy': 0.9565, 'f1': 0.9353, 'roc_auc': 0.9545}
- Random Forest Classifier: {'accuracy': 0.9565, 'f1': 0.9353, 'roc_auc': 0.9545}
- XGBoost Classifier: {'accuracy': 0.9565, 'f1': 0.9353, 'roc_auc': 0.9773}

Industrial Recommendations:
- The selected model relies most on motor_vibration_mm_s, bearing_temperature_c, timestamp_dayofweek. These variabl
- Deploy the Gradient Boosting Classifier pipeline as a baseline for predictive maintenance.
- Validate the selected model on a recent holdout period before connecting it to operator workflows.
- Tune alert thresholds with maintenance teams to balance missed failures and nuisance alarms.